

360 HEATMAP ANALYZER

USER DOCUMENTATION

AUGUST 2021

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Introduction

Thank you for purchasing 360 Heatmap Analyzer. Our team work in any product related to virtual environments and visual communication, like 3d assets, renders, virtual videos, interactives, VR experiences, 360 tours or augmented reality apps. You can find some examples on our website www.sg-digital.cat

Working on data visualization in the videogame industry, we have developed this asset that helps to study, in an intuitive way, user visualization in a 3d scene or 360 video. This asset shows the most visualized screen areas in your scene, using a heatmap 2D image overlapped on the screen.

A heatmap is a data representation technique that measures the magnitude of an interaction through a color code that allows to quickly interpret the results.

With this analysis, you will be able to see which areas are the most visualized, so you can focus on them to study the behavior of users, and the design and user interface of your projects.

We have made the effort so that the result is relative to the time of use of the application, and to avoid applying absolute values that do not depend on the time of use. Thus, if one area of the screen starts to be visualized more than the rest, this area will be updated with the most intense values, while the remaining areas decrease the intensity, a fact that makes the analysis real and dependent on the time of use of each game.

This system causes more calculations to be processed, but the resolution of these calculations can be defined to improve performance, and it also increases the size of the result, which makes it easier to view.

Please remember, it is important for us to have feedback from people like you to improve our products or have a good review in the asset store. Please check out ways that you can directly communicate with us, at the end of this document.

Enjoy the product!

The SG-DIGITAL Team.

Work with 360 Heatmap Analyzer

The asset is prepared to be easy to apply and use in any Unity project.

To set the asset, you first need to put the **sphere_inv** (Fig. 1a) prefab in your scene, that you can locate in “models” folder; **sphere_inv** gameobject must be rotated -270 degrees on X axis and must be at the same position that Main Camera.

You also must assign the material named **materialPixels** (Fig. 1c) located in “Materials” folder that is used to visualize the graphic image. When the scene is not in play mode, the material appears in white color and does not allow to see the scene objects. You can temporarily disable the Image component of the prefab to continue working on your scene in editor mode (Fig. 1d). In play mode, disabling this component allows the application to be used, but the user does not see the analysis result.

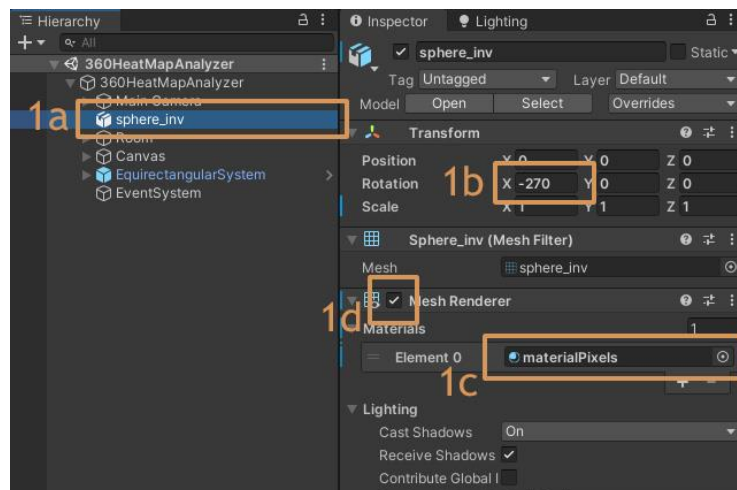


Fig 1. Sphere_inv settings

You have to add the component *heatmapPixelsTime.cs* class to the Main Camera object (Fig.2). In the component dialog, you have to assign the **sphere_inv** gameobject that you put previously to the scene into the Sphere slot (Fig 2a).

The *heatmapPixelsTime.cs* class includes the **ResetValues** method, which allows you to clear the color and reset the analysis. You can also pause the analysis disabling the *heatmapPixelsTime.cs* class on play mode, and continue the analysis enabling the class.

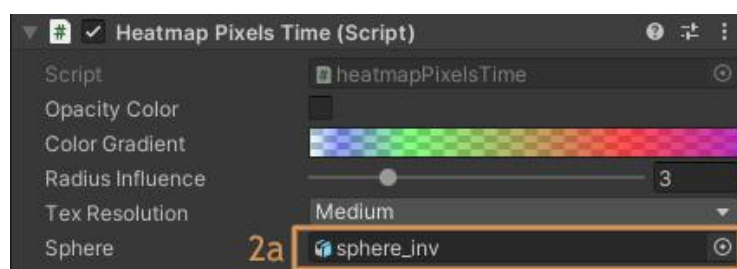


Fig 2. Heatmap Pixels Time dialog

When the application starts, it calculates the time and position of the inputs that the user executes, and creates a heatmap in real time, so it updates the value of the colors based on the total time used each time.

The asset supports all screen resolutions, and you can change the calculation precision to improve the performance. This is especially useful in the case of mobile devices with less processing capacity.

You can see in real time the result of the analysis, or capture a 360 screenshot in equirectangular projection in order to save the visualization of all the data. To enable this option, you have to put the **EquirectangularSystem** prefab, located in “Prefabs” folder, into your scene. This gameobject has a child named **CameraEquirectangular** where you can configure the name, resolution, folder and format of the image, and choose the key to make captures. Finally, you have to assign the *Main Camera* in the **Render From Position** slot (Fig.3).

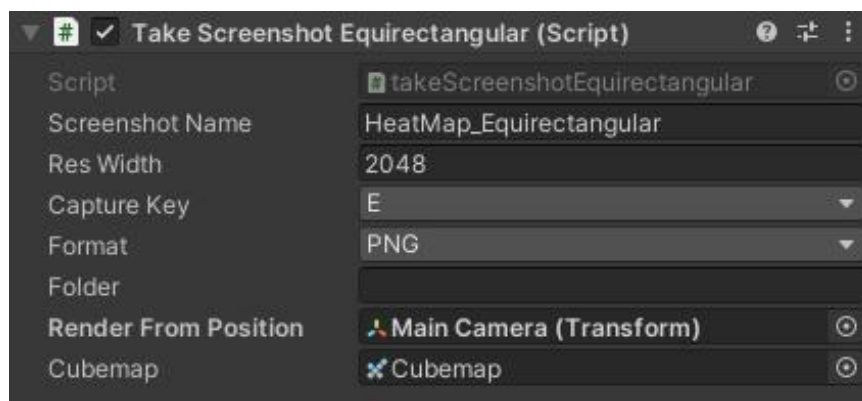


Fig 3. Take Screenshot Equirectangular dialog

When you put the **EquirectangularSystem** prefab into the scene, it's important to disable the Equirectangular layer on the Main Camera (Fig. 4), so that the prefab does not interfere with the visualization of the scene.

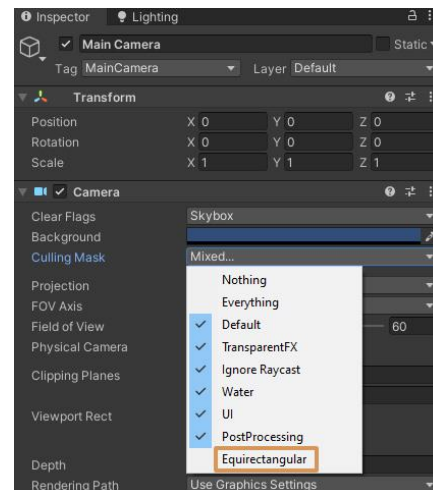


Fig 4. Main Camera dialog

It also includes a scene to test the asset with a room 3d model and the correct configuration of gameobjects. Here, you can explore the possibilities of the asset and configure the best visualization for your project.

Functionalities

The following functionalities can be configured (Fig.5):

- **Opacity Color (Fig.6):** represent the values with a single color with different opacity. More transparent for areas less used and more opaque for the most used areas.
- **Color Gradient (Fig.7):** represent the values with the specified set of colors. This is used only if "Opacity color" is not set. The colors on the left are the colors that represent the areas where the user has less interaction, and those on the right, the areas with more interaction. You can personalize the ramp with your own colors.
- **Radius Influence:** allows modifying the radius of influence of the input on the screen, to have more or less precision of the mouse/ touch.
- **Texture Resolution:** allows modifying the resolution of the 2D image, so the calculation of the values is also simplified, reducing the precision of the representation, but improving the performance (Very high = 100%, High = 50%; Medium = 25%; Low = 12.5%; Very Low=6,25%).
- **Hidden Mode:** allows you to view the result on the screen or just do the calculation in hidden mode, to play without colors and improve performance.
- **Hidden Mode Key:** selectable key for switch hidden mode.

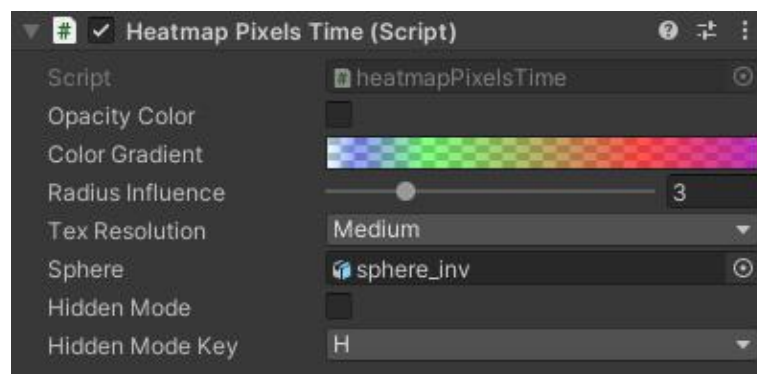


Fig 5. Heatmap Pixels Time dialog

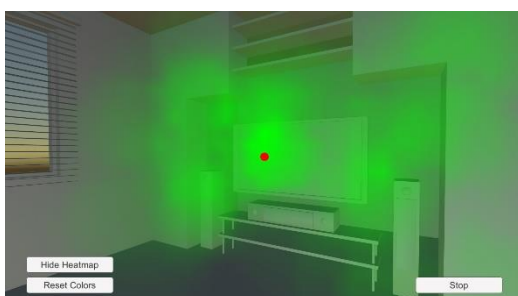


Fig 6. Opacity color mode on

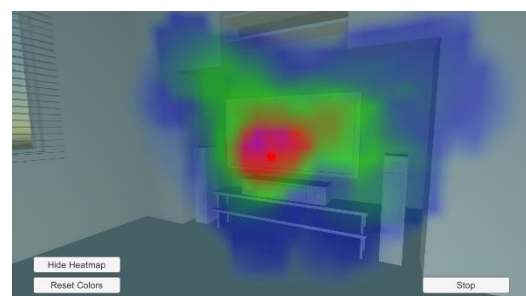


Fig 7. Opacity color mode off

Contact

If you have any problem working with the asset, you can email our support team directly. Our email address is sgdigital.barcelona@gmail.com.

Please note the following when emailing support for quickest service:

1. We need 2-3 days to process email support requests.
2. For fastest service, please provide your Unity Asset Store Purchase invoice number for our product.
3. Put the invoice number in the subject field, with brief description of the problem.
4. In the body of the email, please provide as much of the following as possible, for quickest service:
 - Detailed explanation of the problem. Provide screenshots if possible.
 - It's important to know what version of Unity you use.
 - Sequence of events leading to issue.

Licensing

Licensing for 360 Heatmap Analyzer is governed by the Unity Asset Store End User License Agreement (EULA). For more information about the EULA, please [click here](#) for more details.